RAJALAKSHMI ENGINEERING COLLEGE

Akash M 250701034-2025-CSE

A2

Started on	Friday, 21 August 2026, 4:11 PM
State	Finished
Completed on	Friday, 21 August 2026, 4:11 PM
Time taken	49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void reverse(int n)
{
    int rev = 0, remainder;
    int count = 2;   // initial operations

    while (n != 0)
    {
        count++;     // while condition

        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    count++;         // final while condition

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    reverse(n);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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